

Decoded Bootstrap Paradox V2

— Single Block, Bounded Age

Praveen · Independent Researcher
October 2026

Abstract

This paper proposes a science-fiction resolution to the bootstrap paradox’s infinite-age failure. Information loops, while physical tokens are fresh and have finite worldlines. The model uses a single, unchanging B-theory block. It is a conceptual proposal, not an established physical mechanism.

Problem

In the classic same-watch version, one watch is passed around a closed loop. Treating it as the same object through unlimited repetitions suggests infinite proper age within a finite 40-year interval—a supertask problem often illustrated by Thomson’s lamp. This also confuses recurring information with the physical history of matter and its entropy. Credit: u/ManifoldMold for spotting the supertask issue.

Single-block token diagram

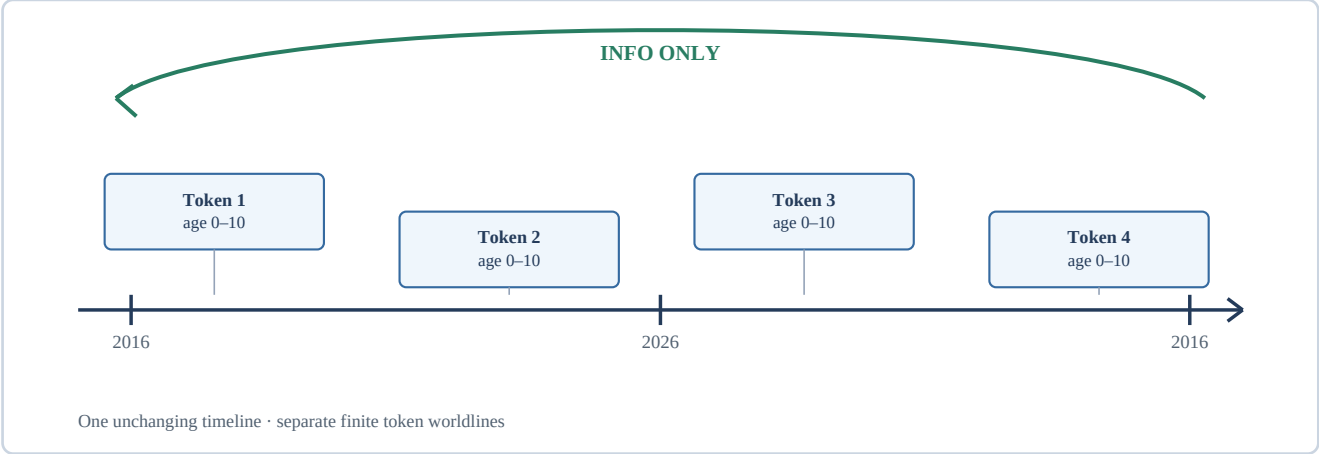


Figure 1. Single block, many tokens.

The loop represents information recurrence only; each token is a distinct physical object.

Rules and Result

Rule 0 — Anchor

The first watch is built normally, without time travel, giving it a defined beginning and end. If the model is mirrored in both directions with no first construction, the origin problem returns. Keep the first build as the anchor.

Rule 1 — Fresh Tokens

Only information loops; matter does not. The design's total information-life is 40 years in the compressed causal history, while each physical token lives normally for 10 years. New matter means a new entropy budget: one design, many tokens. Rejecting same-object identity is intentional (noted by u/SjennyBalaam). Under this model, no entropy violation is claimed.

Rule 2 — Bounded Age

Proper age is never infinite; no Thomson's-lamp limit is required. In the single timeline, only one watch at a time has creation-information participating in the loop—the same object is not aging infinitely. Each token's worldline lasts 10 years.

Result

Single timeline + information-only recurrence + a fresh copy per gap is a realistic science-fiction model. No branching timelines, trees, or multiverse.

V1 → V2 changes

Removed	Branching-tree language
Added	Anchor: first build
Clarified	Proper time versus physical wear

WRONG Same token looping Age → infinity Entropy crossed	RIGHT Fresh tokens Age 0–10 each Entropy ticks per token
---	--

<https://github.com/Praveen-paradox-timeloop/accelerating-decoded-bootstrap-paradox> | [r/timetravel](https://www.reddit.com/r/timetravel) · 5.6K views discussion